AI Procurement Agent

User Manual & Implementation Guide

Platform: n8n Cloud
Primary AI Model: Groq Chat Model
Data Sources: Excel (.xlsx) files in Google Drive and procurement policy documents
Audience: Beginners building and operating the workflow

Purpose of this manual

This manual explains how to build, configure, test, operate, troubleshoot, and document an AI Procurement Agent in n8n Cloud. It follows the workflow developed for the project and is written as a practical reference rather than only a conceptual overview.

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1. Project Overview

Project name: AI Procurement Agent

Core idea: automate the repetitive parts of procurement while keeping the final purchasing decision under human control.

The agent receives an employee procurement request, understands the requirement, checks available inventory, verifies procurement policy, identifies approved suppliers, collects or reads quotations, compares supplier offers, prepares a recommendation, obtains human approval, generates a purchase order for approved requests, and records important actions in an audit log.

> The system is designed as an **assistive agent**. The AI does not independently authorize purchases. The human approval step is the control point before a purchase order is generated.

1.1 Business Problem

Traditional procurement workflows can involve email exchanges, spreadsheet lookups, manual supplier checks, policy interpretation, quotation comparison, and repeated status updates. These activities consume time and can create inconsistent processing or incomplete documentation.

- Employees need a simple way to submit procurement requirements.
- Procurement teams need to verify whether requested items are already available.
- Requests must comply with procurement policies and approval rules.
- Only approved suppliers should be considered.
- Multiple quotations may need to be compared using price, delivery, warranty, rating, and payment terms.
- Human approvers need a clear summary before deciding.
- Important actions need an auditable record.

1.2 Proposed Solution

The AI Procurement Agent orchestrates these activities in n8n. Structured information comes from Excel files and other business documents. AI is used for requirement understanding, structured extraction, supplier comparison, and purchase-order generation. Deterministic checks handle inventory, policy, and supplier approval conditions. A human makes the final approval decision.

2. Scope, Stakeholders, and Outcomes

Typical stakeholders include employees requesting equipment or services, procurement officers, procurement managers/approvers, finance teams, approved suppliers, and audit/compliance teams.

Stakeholder	Need from the system
Employee / Requester	Submit a clear requirement and receive status visibility.
Procurement Team	Reduce manual lookup, policy checking, and quotation comparison.
Approver / Manager	Review a structured recommendation and make the final decision.
Finance	Receive consistent purchase-order information and an auditable trail.
Supplier	Receive accurate purchase-order information after approval.
Audit / Compliance	Maintain traceable records of actions, decisions, and status.

Expected outcomes include reduced repetitive work, consistent checks, faster processing, better traceability, and a clear separation between AI recommendations and human authorization.

3. Solution Architecture

High-level flow

```
Employee Request → AI Requirement Understanding → Specification Extraction → Inventory Check → Procurement Policy Check → Approved Supplier Search → Quotation Collection → Supplier Comparison → Recommendation → Human Approval → Purchase Order → Audit & Reporting
```

Architecture layers

Layer	Responsibility
Input	Read employee request and reference data.
Understanding	Use AI to understand the request and extract specifications.
Business Rules	Check inventory, procurement policy, and supplier approval.
Supplier Analysis	Collect/prepare quotations and compare supplier offers.
Decision Control	Present recommendation to a human and branch on approval.
Execution	Generate a purchase order after approval.
Audit	Append the outcome to the existing audit log XLSX.

The architecture deliberately combines AI with deterministic workflow nodes. AI is useful for language-heavy tasks; rules and structured data are preferable for exact compliance checks.

4. Prerequisites

- An n8n Cloud account.
- A connected Google Drive account containing the project Excel files.
- A working Groq API credential configured in n8n for the Groq Chat Model.
- Employee request, inventory, approved-supplier, quotation, and audit-log Excel files.
- Procurement policy documentation in a readable document/PDF form if policy retrieval is part of the workflow.
- A defined human approval mechanism in n8n.
- Test data that does not contain sensitive production information.

Recommended naming convention

File	Purpose
employee-requests.xlsx	Employee procurement requests.
inventory.xlsx	Current inventory and stock information.
approval_matrix(1).xlsx	Approved supplier master / approval matrix.
supplier-quotations.xlsx	Supplier quotation records.
Audit Log.xlsx	Persistent audit history.
procurement-policy.pdf	Policy and procurement rules.

5. Input Excel Files and Data Design

5.1 Employee Request

Suggested column	Example	Purpose
request_id	R001	Unique request identifier.
employee_id	E101	Requester identifier.
product	Laptop	Requested item.
category	IT Equipment	Procurement category.
quantity	10	Requested quantity.
required_date	2026-10-10	Needed-by date.
specifications	16GB RAM, 512GB SSD	Technical requirement.

5.2 Inventory

The inventory file should contain enough information to determine whether the requested product is already available, such as product/item name, category, available quantity, and optionally location or stock status.

5.3 Approved Supplier Matrix

Column	Role
supplier_record_id	Unique supplier record.
supplier_id	Supplier identifier.
supplier_name	Supplier name.
category	Supplier's approved category.
products	Products supplied.
contact_email	Quotation/contact communication.
phone	Optional contact number.
approved_status	Approved / Not Approved.
supplier_rating	Supplier rating.
payment_terms	Payment terms, if maintained here.
location	Supplier location, if required.

5.4 Supplier Quotations

Column	Role
quotation_id	Quotation identifier.
request_id	Links quotation to request.
supplier_id	Links quotation to supplier.
product	Quoted product.
quantity	Quoted quantity.
unit_price	Price per unit.

Column	Role
total_price	Total quotation amount.
delivery_days	Delivery lead time.
warranty_years	Warranty duration.
quotation_date	Date quotation was created.
valid_until	Quotation validity.
payment_terms	Payment terms.
quotation_status	Quotation status.

5.5 Audit Log

The project's existing audit log has exactly these columns:

```
audit_id, request_id, timestamp, agent_step, action, input_data, output_data, decision,
performed_by, status
```

6. n8n Workflow Construction

6.1 Start with the Employee Request

1. Create a Manual Trigger for initial development/testing.
2. Add Google Drive and select File → Download.
3. Select employee-requests.xlsx.
4. Add Extract From File after the download.
5. Configure the extractor for XLSX and use the binary property produced by Google Drive, commonly named data.
6. Execute the extraction and confirm that each spreadsheet row appears as an n8n item.

Important beginner rule: never feed the downloaded binary XLSX directly into an AI Agent. Convert/extract the file into structured JSON first. This avoids binary-data errors in agent inputs.

6.2 Inventory Branch

1. Download inventory.xlsx from Google Drive.
2. Extract the XLSX into JSON.
3. Merge the request data and inventory data in a way that preserves the fields required by the next business-rule step.
4. Check requested product and required quantity against available inventory.
5. Record the inventory result clearly, such as Available or Not Available.

6.3 Procurement Policy Check

The policy-check stage should determine whether the request satisfies applicable procurement rules. Where policy text is supplied to an AI Agent, instruct the agent to use only the supplied policy and return a structured result. Where a rule can be evaluated deterministically, prefer an IF/Switch or other rule node.

Typical output fields:

```
policy_status = Approved / Rejected
policy_reason = explanation
request_id = request identifier
```

6.4 IF Node — Policy Result

Configure the IF node using the actual JSON field produced by the policy-check step. Example:

```
Value 1: {{$json.policy_status}}
Operation: is equal to
Value 2: Approved
```

TRUE continues to supplier processing. FALSE should stop/reject the procurement path or route it to an appropriate review/audit path.

7. Approved Supplier Search and Quotation Processing

7.1 Approved Supplier Filter

The supplier filter should ensure that the supplier is approved and relevant to the requested category/product.

Typical checks:

- approved_status = Approved
- supplier category matches the request category
- supplier products contain or correspond to the requested product

> Use actual field names from the extracted supplier JSON. Do not assume that a field is named exactly as shown if the spreadsheet uses a different header.

7.2 Quotation Data

After approved suppliers are identified, quotation records can be filtered by request_id and supplier_id. The quotation information should be structured before being sent to an AI comparison agent.

7.3 Supplier Comparison AI Agent

The comparison agent should compare suppliers using available evidence such as total price, delivery days, warranty years, supplier rating, and payment terms. It must not invent missing values.

Recommended system instruction:

```
You are an AI Procurement Supplier Comparison Agent. Compare only quotations and supplier
information provided in the input. Consider total price, delivery days, warranty, supplier
rating, and payment terms. Do not invent missing information. Produce a transparent comparison
and recommendation. The recommendation is not the final purchasing decision; a human approver
must decide.
```

> If multiple supplier rows arrive separately, aggregate them before the AI Agent so the model can see all relevant quotations in one input.

8. Human Approval and Decision Branch

The Human Approval step is the control point between recommendation and purchase-order generation.

Recommended approval fields:

Field	Values / purpose
procurement_decision	Approve or Reject.
approval_comments	Human explanation or comments.

After the approval node, add an IF node named **Approval Decision**.

Example configuration:

```
Value 1: {{$json.procurement_decision}}
Operation: is equal to
Value 2: Approve
```

TRUE → Generate PO. FALSE → Rejected Audit.

> Always inspect the Human Approval node's Output → JSON before creating the expression. If the actual decision field is not procurement_decision, use the real field name.

9. Purchase Order Generation

9.1 TRUE Branch

Connect the TRUE output of Approval Decision to an AI Agent named **Generate PO**.

```
Suggested system message:
```

```
You are an AI Procurement Purchase Order Generator. Generate a purchase order using only the approved procurement information supplied in the input. Include Request ID, Supplier ID, Supplier Name, Product, Quantity, Unit Price, Total Price, Delivery Days, Warranty Years, Payment Terms, and Purchase Order Number. Do not invent or modify supplier, quotation, pricing, or quantity information. The request has already received human approval.
```

The PO agent should use structured source fields whenever possible. A generated PO should be treated as an output document/record, not as a substitute for the human approval decision.

9.2 Approved Audit Record

After Generate PO, add Edit Fields and name it **Prepare Approved Audit**. Add exactly the same 10 columns as your audit file.

Audit field	Approved-branch value
audit_id	AUD-{{$now.toMillis()}}
request_id	Actual request_id expression from JSON
timestamp	{{$now}}
agent_step	Approval Decision
action	Procurement Request Approved
input_data	Request/product/quantity or other relevant input summary
output_data	AI PO output / PO generation result
decision	Approved
performed_by	Human
status	Completed

Turn on **Include Other Input Fields** when appropriate so useful incoming fields are not unnecessarily discarded.

9.3 FALSE Branch — Rejected Audit

Connect FALSE to Edit Fields named **Prepare Rejected Audit**.

Audit field	Rejected-branch value
audit_id	AUD-{{$now.toMillis()}}
request_id	Actual request_id expression from JSON
timestamp	{{$now}}
agent_step	Approval Decision
action	Procurement Request Rejected
input_data	Request/product/quantity or relevant approval input
output_data	Procurement request rejected by human approval
decision	Rejected
performed_by	Human
status	Completed

10. Audit Logging to the Existing XLSX File

This project uses an existing Audit Log.xlsx file stored in Google Drive. Because the file is XLSX rather than a native Google Sheet, the workflow must read the existing workbook, append the new audit record, recreate the XLSX, and update the existing Drive file.

10.1 Logical Architecture

```
Approved Audit + Rejected Audit → Merge #1 (Append) → New Audit Record
Existing Audit Log.xlsx → Google Drive Download → Extract From File
New Audit Record + Existing Audit Records → Merge #2 (Append) → Convert to XLSX → Google Drive Update
```

10.2 Merge #1

Merge #1 is intended to consolidate the approved/rejected audit path into the new audit-record stream. Use Append when the goal is to keep all incoming items rather than match records by a key.

10.3 Download the Existing Audit Log

1. Add Google Drive after the point where the audit-log processing is initiated.
2. Resource: File.
3. Operation: Download.
4. Select the existing Audit Log.xlsx.
5. Execute the node and confirm the file appears as binary data, commonly under data.

10.4 Extract the Existing Audit Log

1. Add Extract From File after Download.
2. Choose the XLSX extraction option available in your n8n version.
3. Set the input binary field to the binary property produced by the Download node, commonly data.

4. Execute the node.
5. Confirm that the existing spreadsheet rows appear as JSON items.

10.5 Merge #2

Merge #2 combines the existing audit records with the new audit record.

```
Input 1 → Extract From File (existing audit rows)
Input 2 → Merge #1 (new audit record)
Mode → Append
```

> Do not configure matching fields for this append operation. If the existing workbook contains 100 rows and one new audit record is generated, the output should contain 101 items.

10.6 Convert Back to XLSX

1. Add Convert to File (or the spreadsheet/XLSX conversion node available in your n8n version) after Merge #2.
2. Convert JSON items to XLSX.
3. Confirm that the output is binary data and that the binary property is commonly named data.

10.7 Update the Existing Google Drive File

> Add Google Drive after the XLSX conversion. If your n8n version exposes an Update operation for an existing file, select File → Update, select the existing Audit Log.xlsx, and specify the binary property containing the newly created XLSX. Do not use Upload blindly if you intend to replace the existing workbook; uploading may create a second file instead of updating the original.

Always test the final Drive step in a test folder or with a backup copy before using the workflow against an important audit file.

11. Testing Strategy

Test the workflow in small sections before executing the complete workflow.

Test	Expected result
Employee XLSX download	File downloads successfully.
Employee XLSX extraction	Rows appear as JSON items.
Inventory check	Correct available/not-available result.
Policy check	Correct policy status and reason.
Supplier filter	Only approved/relevant suppliers continue.
Quotation comparison	All relevant quotations are visible to the AI.
Human approval — Approve	TRUE branch executes and PO is generated.
Human approval — Reject	FALSE branch executes and rejection audit is created.
Audit Merge #2	Existing rows plus new row are present.
XLSX conversion	Binary XLSX is produced.
Drive update	Existing audit workbook is updated without unwanted duplicates.

Recommended approval test cases

- Approve a valid request with multiple approved supplier quotations.

- Reject the same request and confirm no PO is generated.
- Use a request whose product is already sufficiently available in inventory.
- Use a request that fails the policy check.
- Use a supplier marked Not Approved and confirm it does not reach the recommendation stage.
- Use missing optional quotation information and confirm the AI does not invent it.

12. Troubleshooting Guide

Problem	Likely cause	Fix
Agent says binary data is unsupported	XLSX binary was sent directly to AI Agent.	Use Google Drive Download → Extract From File → AI Agent.
Expression returns empty	Field name differs from assumed name.	Open previous node → Output → JSON and use the actual field.
Merge has wrong number of items	Wrong mode or input connection.	Use Append and verify Input 1/2 connections.
Old audit rows disappear	Only the new record was converted to XLSX.	Download/extract old log and append it in Merge #2.
Audit Log (1).xlsx appears	Upload created a new file.	Use an existing-file Update operation or carefully replace the origina
AI invents supplier information	Prompt allows unsupported inference.	Explicitly require use of supplied data only and prohibit invention.
Human decision branch doesn't work	Incorrect IF field/expression.	Inspect Human Approval JSON and use the actual decision field.
No quotations are compared	Quotation rows arrive separately or are filtered incorrectly.	Aggregate/filter by request_id and supplier_id before comparison.
PO generated after rejection	TRUE/FALSE condition is wrong.	Test Approval Decision independently with both decision values.

Debugging method

1. Execute only the failing node.
2. Open Input → JSON.
3. Confirm field names and values.
4. Check whether the node received the expected number of items.
5. Correct the expression or node configuration.
6. Execute again before reconnecting downstream nodes.

13. Security, Governance, and Operational Controls

- Keep API credentials in n8n credentials; do not hard-code keys in prompts or spreadsheets.
- Use test/sample data while developing the workflow.
- Limit Google Drive permissions to the files/folders required by the workflow.
- Keep human approval before any purchase-order generation that commits the organization to a purchase.
- Do not let an AI model invent prices, suppliers, approval statuses, or policy rules.
- Store enough audit information to reconstruct what happened, while avoiding unnecessary sensitive data in input_data/output_data.
- Back up the audit workbook before changing the update mechanism.
- Review AI prompts and business rules when procurement policy changes.

Human-in-the-loop principle

> The AI can summarize, compare, classify, and generate structured outputs, but the procurement authorization remains a human decision. This is especially important where purchases have financial, compliance, or supplier-risk consequences.

14. Deployment and Maintenance Checklist

Before production use:

- Replace Manual Trigger with the desired production trigger or intake mechanism.
- Verify all Google Drive credentials and file permissions.
- Verify Groq credentials and model availability.
- Confirm every expression against real node JSON.
- Test both approval outcomes.
- Back up Audit Log.xlsx.
- Confirm the Google Drive operation updates the intended file rather than creating duplicates.
- Document the procurement policy version used by the workflow.
- Add error handling/notifications for failed downloads, AI errors, missing quotations, and failed audit updates.
- Monitor workflow executions and review failures.

Maintenance

- Update supplier master data when supplier approval status changes.
- Update quotation schemas if the procurement process changes.
- Review AI prompts when business requirements change.
- Review the audit schema before adding/removing columns.
- Retest the workflow after n8n node-version or credential changes.

15. Final End-to-End Checklist

#	Checkpoint
1	Employee request XLSX can be downloaded and extracted.
2	AI requirement understanding returns structured information.
3	Inventory is checked against the request.
4	Procurement policy is checked.
5	Only approved/relevant suppliers continue.
6	Quotation data is available for comparison.
7	Supplier Comparison AI Agent produces a transparent recommendation.
8	Human Approval collects Approve/Reject and comments.
9	TRUE branch generates a PO only after approval.

#	Checkpoint
10	FALSE branch creates a rejection audit record.
11	Approved/rejected audit output is consolidated.
12	Existing Audit Log.xlsx is downloaded and extracted.
13	Merge #2 uses Append to combine old and new audit records.
14	The complete dataset is converted back to XLSX.
15	The existing Google Drive audit file is updated safely.
16	Both approval outcomes have been tested.

Project completion criterion

The project is ready for demonstration when a test procurement request can travel from request intake through requirement understanding, inventory/policy/supplier checks, quotation comparison, human approval, PO generation or rejection, and final audit logging without losing the original audit history.

Appendix A — Recommended Node Naming

Stage	Suggested node name
Trigger	Manual Trigger
Request file	Download Employee Requests
Request extraction	Extract Employee Requests
Inventory	Download Inventory / Extract Inventory
Requirement AI	AI Requirement Understanding
Policy	Procurement Policy Check
Policy branch	Policy Decision
Supplier master	Download Approved Suppliers / Extract Suppliers
Supplier filter	Approved Supplier Filter
Quotation	Download Quotations / Extract Quotations
Comparison	Supplier Comparison AI Agent
Approval	Human Approval
Approval branch	Approval Decision
PO	Generate PO
Approved audit	Prepare Approved Audit
Rejected audit	Prepare Rejected Audit
Audit consolidation	Merge #1 / Merge #2
Audit extraction	Extract Audit Log
Excel creation	Create Updated Audit XLSX
Drive update	Update Audit Log

Appendix B — Useful n8n Expression Examples

Unique audit ID:

```
AUD-{{$now.toMillis()}}
```

Current timestamp:

```
{{$now}}
```

Example IF decision:

```
Value 1: {{$json.procurement_decision}}
Operation: is equal to
Value 2: Approve
```

Important: these examples are templates. Always verify the actual field names in the previous node's JSON output before using an expression.